ST117 Pod FINAL Written Report - Phase 2 Task 4 (Fine grains - Butterflies)

Report Pod MS-05

2026-05-09

Contributions

This submission was created by: 1. u5692931: a, b, c, d

a)

include an output of three rows

```
vf <- read.csv("Vegetation/VF_FinGrain_Vegetation_1994-2015/Raw_Data/ECN_VF1_RawData.csv")
ib <- read.csv("Invertebrates/Lepidoptera_Butterflies_1993-2015/Raw_Data/ECN_IB1_RawData.csv")

ib$YEAR <- year(dmy(ib$SDATE))

vf_veg        <- vf[vf$FIELDNAME == "VEG_SPEC", ] # ruling out non-"veg-spec"

vf_spec_count <- summarise(group_by(vf_veg, SITECODE, SYEAR),
                           finegrain_species = n_distinct(VALUE),
                           .groups = "drop") # one row per combo

colnames(vf_spec_count)[colnames(vf_spec_count) == "SYEAR"] <- "YEAR" # rename for consistency

# Same thing for ib
ib_clean      <- ib[ib$FIELDNAME != "XX", ]

ib_count      <- summarise(group_by(ib_clean, SITECODE, YEAR),
                           total_butterflies = sum(VALUE, na.rm = TRUE),
                           .groups = "drop")

combined  <- merge(vf_spec_count, ib_count, by = c("SITECODE", "YEAR")) # merging for output

kable_styling(
  kable(combined[1:3, ],
        caption   = "First three rows of the aggregated
        Fine Grain species and Butterfly count dataset",
        longtable = FALSE,
        format    = "latex"),
  latex_options = c("scale_down", "hold_position")
)
```

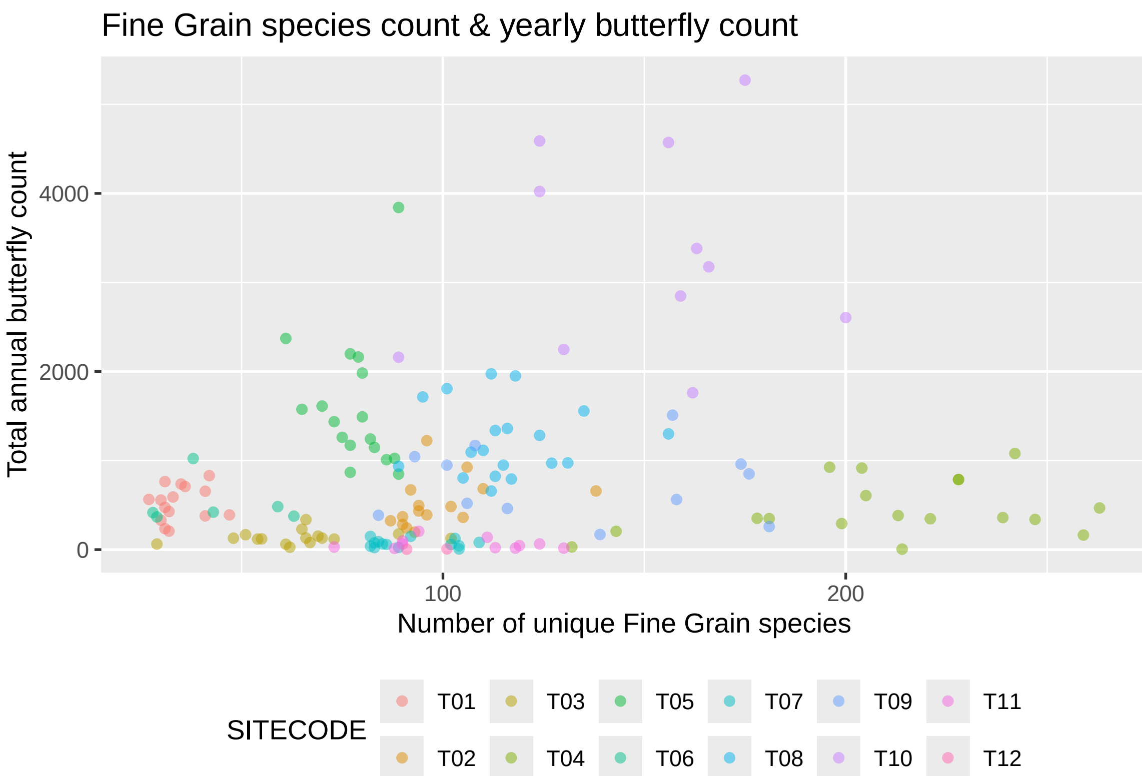
Table 1: First three rows of the aggregated Fine Grain species and Butterfly count dataset

SITECODE	YEAR	finegrain_species	total_butterflies
T01	1994	32	428
T01	1996	35	737
T01	1997	42	831

b)

Limit: 100 words + 1 figure

```r
ggplot(combined, aes(x = finegrain_species, y = total_butterflies, colour = SITECODE)) +
  geom_point(alpha = 0.5, size = 1.5) +
  labs(x = "Number of unique Fine Grain species",
       y = "Total annual butterfly count",
       title = "Fine Grain species count & yearly butterfly count") +
  theme(legend.position = "bottom") +
  guides(colour = guide_legend(nrow = 2))
```



Within the same time period, butterfly abundance and fine grain plant species richness are expected to correlate positively, since more plant variety provides richer food resources, supporting greater butterfly populations. However, the scatterplot reveals nearly no such relationship, whether linear or otherwise. Instead, points of the same site colour cluster closely together, suggesting that butterfly counts remain consistent within each site regardless of vegetation richness. This implies that site-level conditions, such as climate, temperature and humidity, are the determining factors of butterfly abundance. Furthermore, several outliers are observed in the butterfly count axis, one of which is notably extreme.

c)

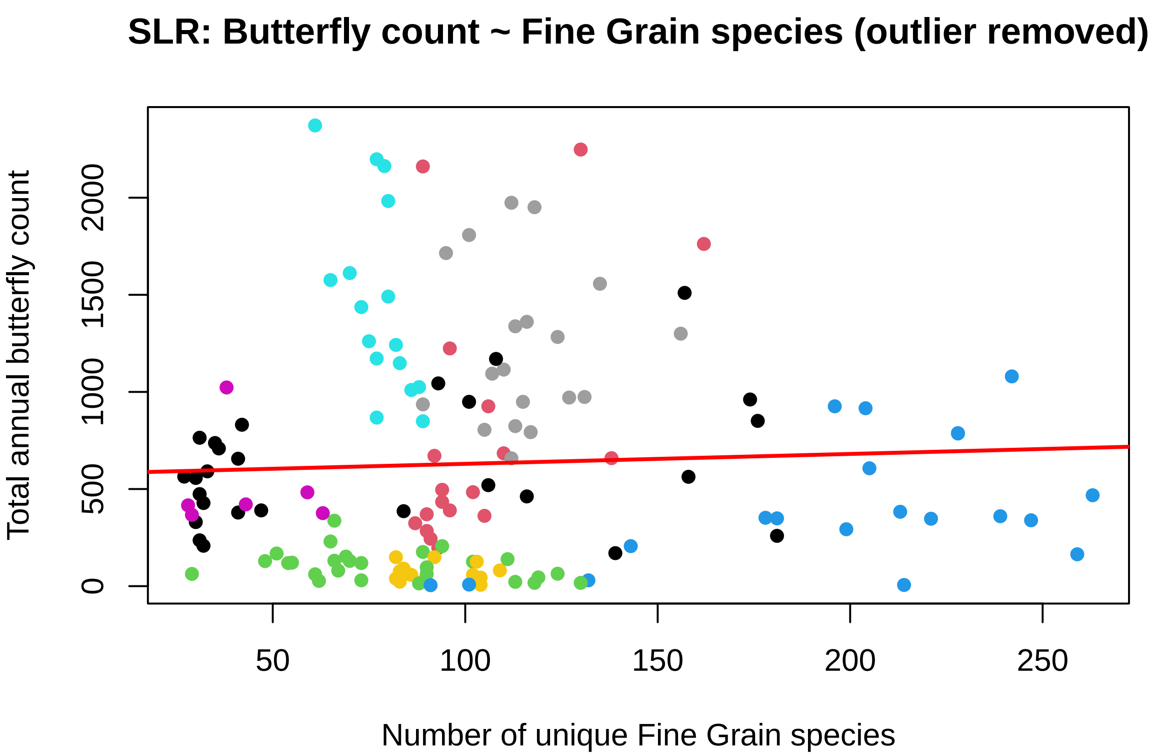
Limit: 200 words + 5 figures + 1 table

To best model linear regression, we need to remove outliers. Outliers beyond 1.5×IQR from Q1 and Q3 will be removed from the regression dataset.

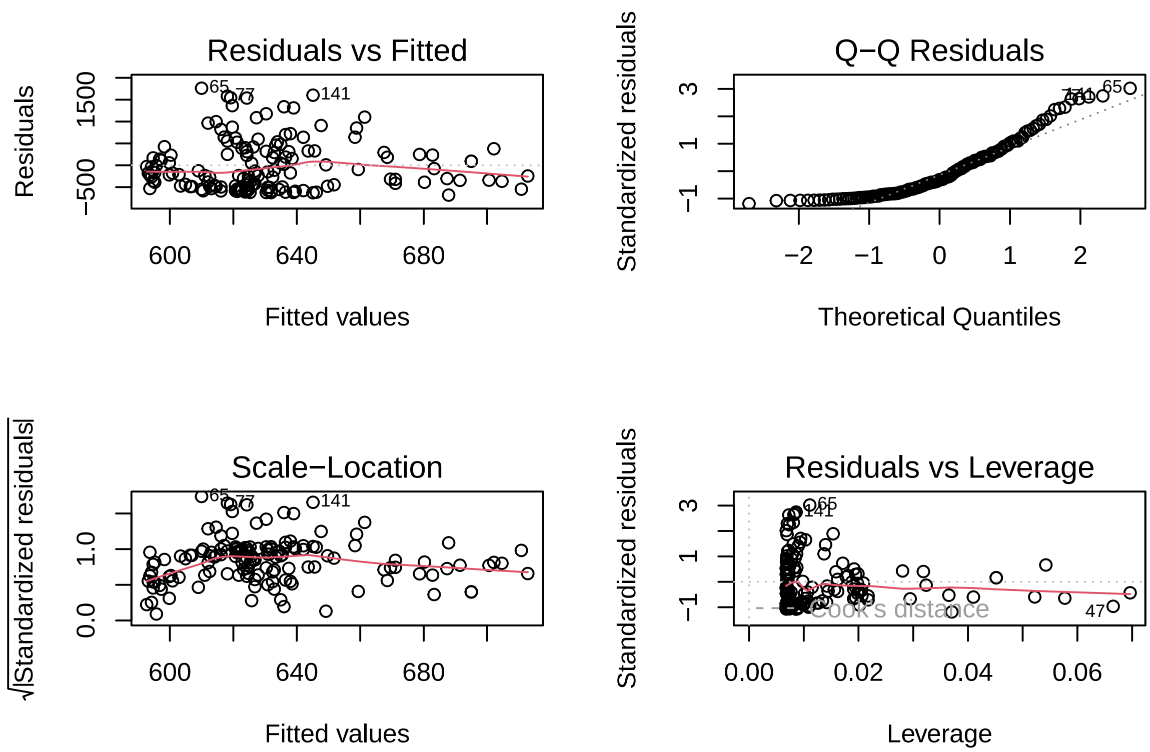
```r
# Remove outliers
threshold       <- quantile(combined$total_butterflies, 0.75) +
  1.5 * IQR(combined$total_butterflies)
#upper outlier threshold
remain <- combined[combined$total_butterflies <= threshold, ] #rule out outliers

# Fit model
lin_mod <- lm(total_butterflies ~ finegrain_species, data = remain)
#fitting linear model in order of response ~ predictor
all_para  <- summary(lin_mod) #store all the parameters
```

```r
# Regression line on scatter plot
plot(remain$finegrain_species, remain$total_butterflies,
     col  = as.factor(remain$SITECODE),
     pch  = 16,
     xlab = "Number of unique Fine Grain species",
     ylab = "Total annual butterfly count",
     main = "SLR: Butterfly count ~ Fine Grain species (outlier removed)")
abline(lin_mod, col = "red", lwd = 2)
```




```r
# Parameters plotting
par(mfrow = c(2, 2)) # having 2 each row
plot(lin_mod)
```



```r
par(mfrow = c(1, 1)) #reset back to 1

# Table: coefficient summary
all_coefs <- data.frame(
  Term     = c("Intercept", "finegrain_species"),
  Estimate = round(coef(lin_mod), 3),
  SE       = round(all_para$coefficients[, 2], 3),
  t        = round(all_para$coefficients[, 3], 3),
  p        = formatC(all_para$coefficients[, 4], format = "e", digits = 2),
  row.names = NULL
)

kable_styling(
  kable(all_coefs,
        col.names = c("Term", "Estimate", "Std. Error", "t value", "Pr(>|t|)"),
        caption   = paste0("SLR: total\\_butterflies ~ finegrain\\_species  (n = ",
                           nrow(remain),
                           ", R² = ", round(all_para$r.squared, 3), ")"),
        longtable = FALSE,
        format    = "latex"),
  latex_options = c("hold_position")
)
```

Table 2: SLR: total_butterflies finegrain_species (n = 147, R^2 = 0.002)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	578.950	106.125	5.455	2.06e-07
finegrain_species	0.509	0.918	0.554	5.80e-01

After removal of outliers, a linear regression line is plot, taking `finegrain_species` as independent variable and butterfly count as dependent. We can clearly see that the line represents only mild positive regression (R^2 is extremely low), which indicates that this relationship is only a minor determining factor. The four diagnostic plots reveal more insight. The plot shows a "fan" pattern where the points spread out more as the fitted values increase. In the QQ plot, most points follow the diagonal line, but at the higher end, the points curve downwards and drift away from the path. This indicates a "light-tailed" distribution. The red line in the standardised residuals plot shows a slight upward trend, and the standardized residuals spread out more as the fitted values increase. This implies that the uncertainty in data prediction is not stable and as the model predicts higher butterfly counts, the actual data points become more scattered. The leverage plot shows influential points remain within safe limits. Overall, the model captures a real but weak biological trend.

d)

Limit: 100 words + 12 figures + 12 tables

```r
sites <- unique(remain$SITECODE) #grab unique sites

for (s in sites) { #each value within T01 to T12
  site_data <- remain[remain$SITECODE == s, ]

  lm_site  <- lm(total_butterflies ~ finegrain_species, data = site_data) #same linear model
  s_site   <- summary(lm_site)

  plot(site_data$finegrain_species, site_data$total_butterflies,
       col  = "darkblue",
       pch  = 16,
       xlab = "Number of unique Fine Grain species",
       ylab = "Total annual butterfly count",
       main = paste("Site", s, ": Butterfly count ~ Fine Grain species"))
  abline(lm_site, col = "red", lwd = 2)
```

```r
  coef_df <- data.frame(
    Term      = c("Intercept", "finegrain_species"),
    Estimate = round(coef(lm_site), 3),
    SE        = round(s_site$coefficients[, 2], 3),
    t         = round(s_site$coefficients[, 3], 3),
    p         = formatC(s_site$coefficients[, 4], format = "e", digits = 2),
    row.names = NULL
  )

  print(kable_styling(
    kable(coef_df,
          col.names = c("Term", "Estimate", "Std. Error", "t value", "Pr(>|t|)"),
          caption   = paste0("Site ", s, "  (n=", nrow(site_data),
                             ", R²=", round(s_site$r.squared, 3), ")"),
          longtable = FALSE,
          format    = "latex"),
    latex_options = c("hold_position")
  ))
}
```

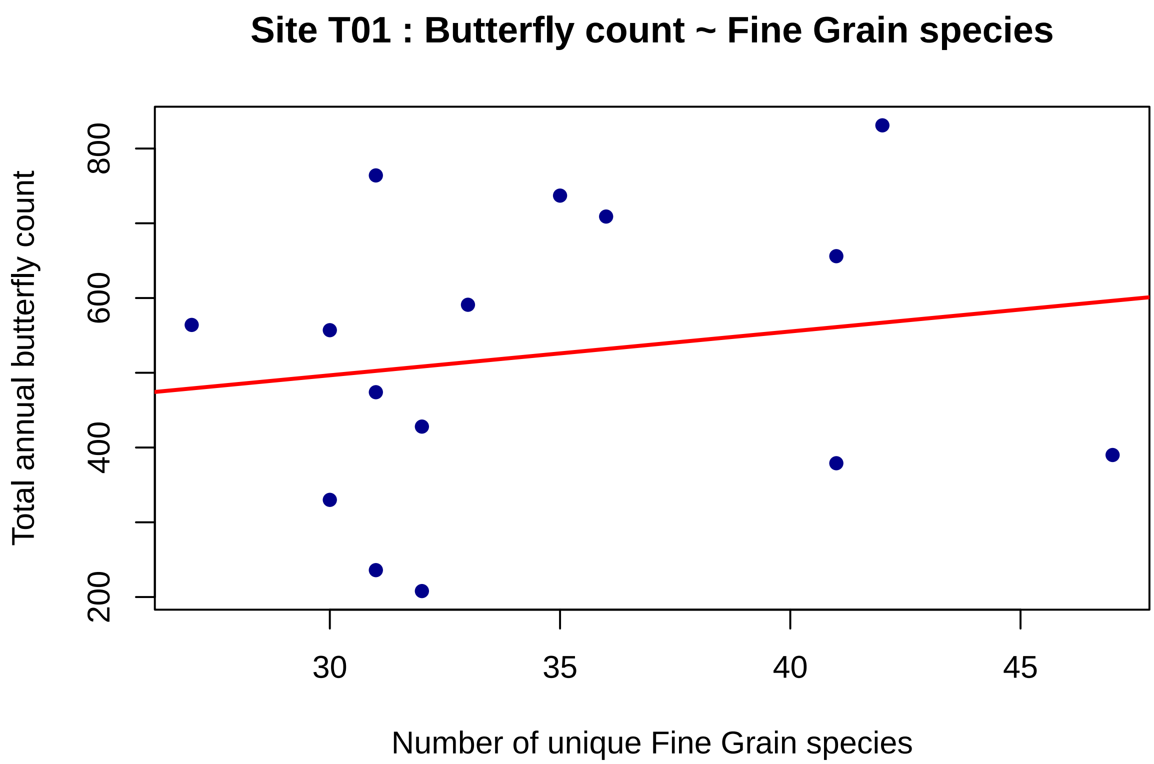


Table 3: Site T01 (n=15, R^2=0.029)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	320.715	328.511	0.976	3.47e-01
finegrain_species	5.864	9.378	0.625	5.43e-01

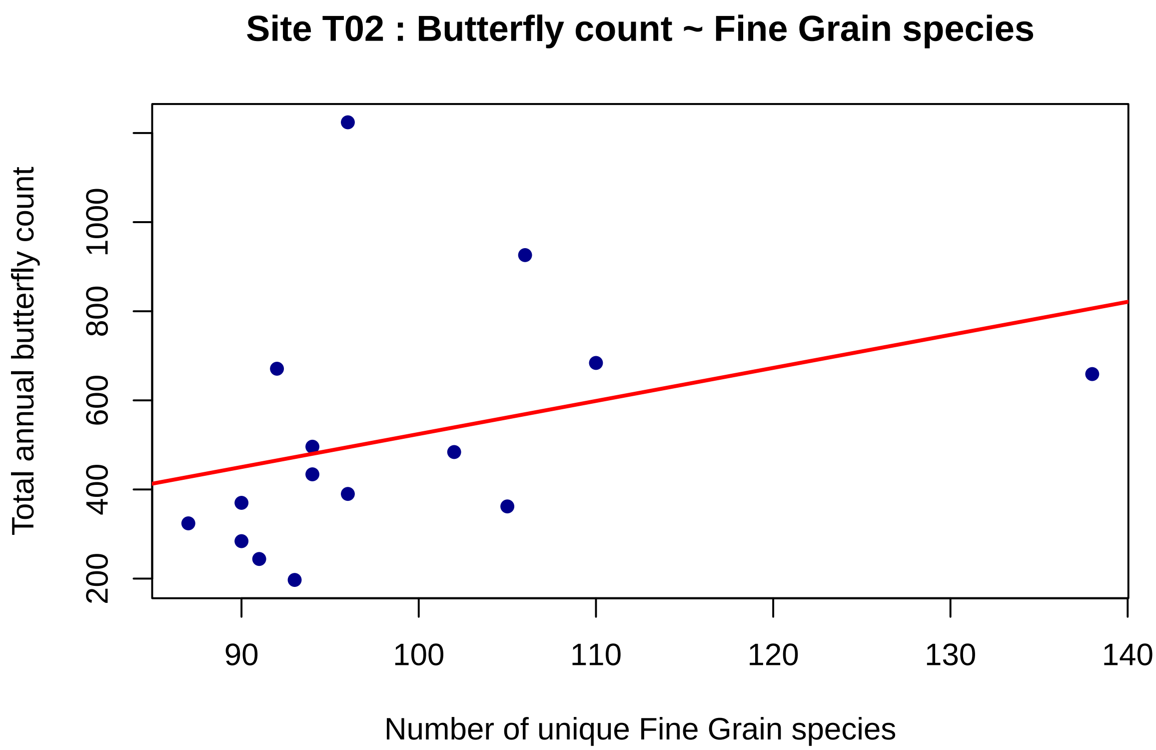


Table 4: Site T02 (n=15, R^2=0.115)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	-217.017	568.012	-0.382	7.09e-01
finegrain_species	7.415	5.698	1.301	2.16e-01

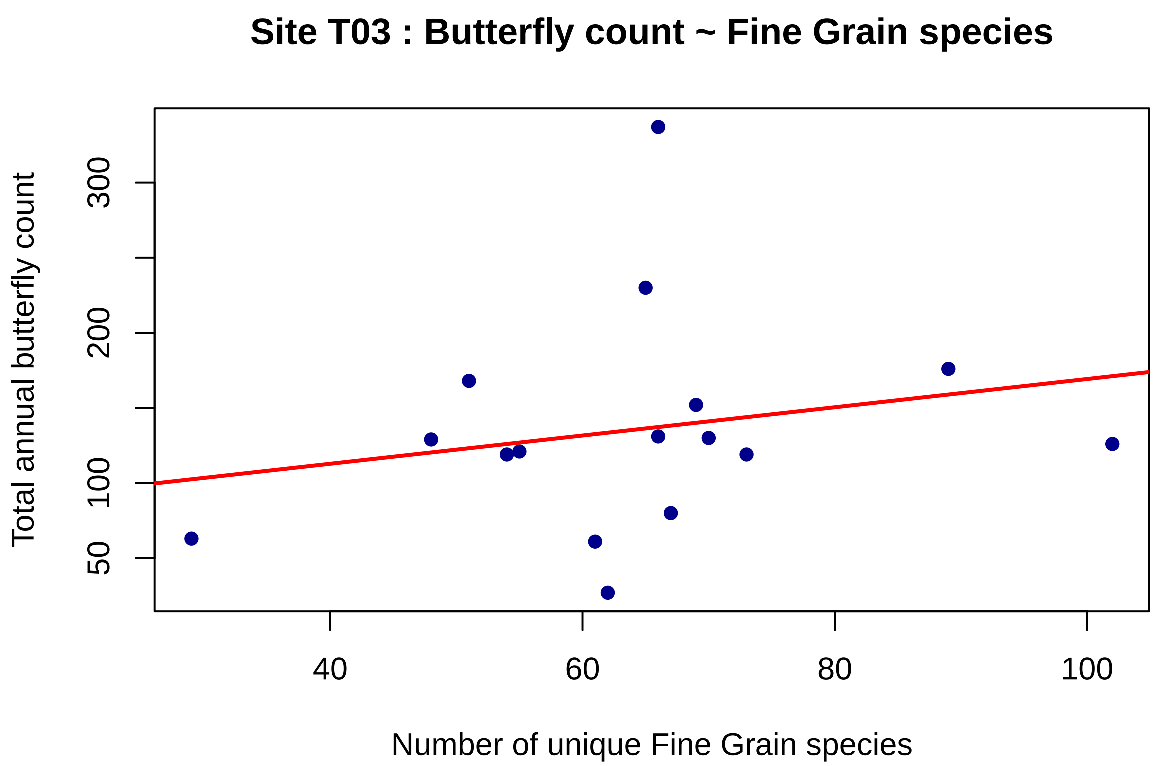


Table 5: Site T03 (n=16, R^2=0.046)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	75.207	76.110	0.988	3.40e-01
finegrain_species	0.940	1.151	0.817	4.28e-01

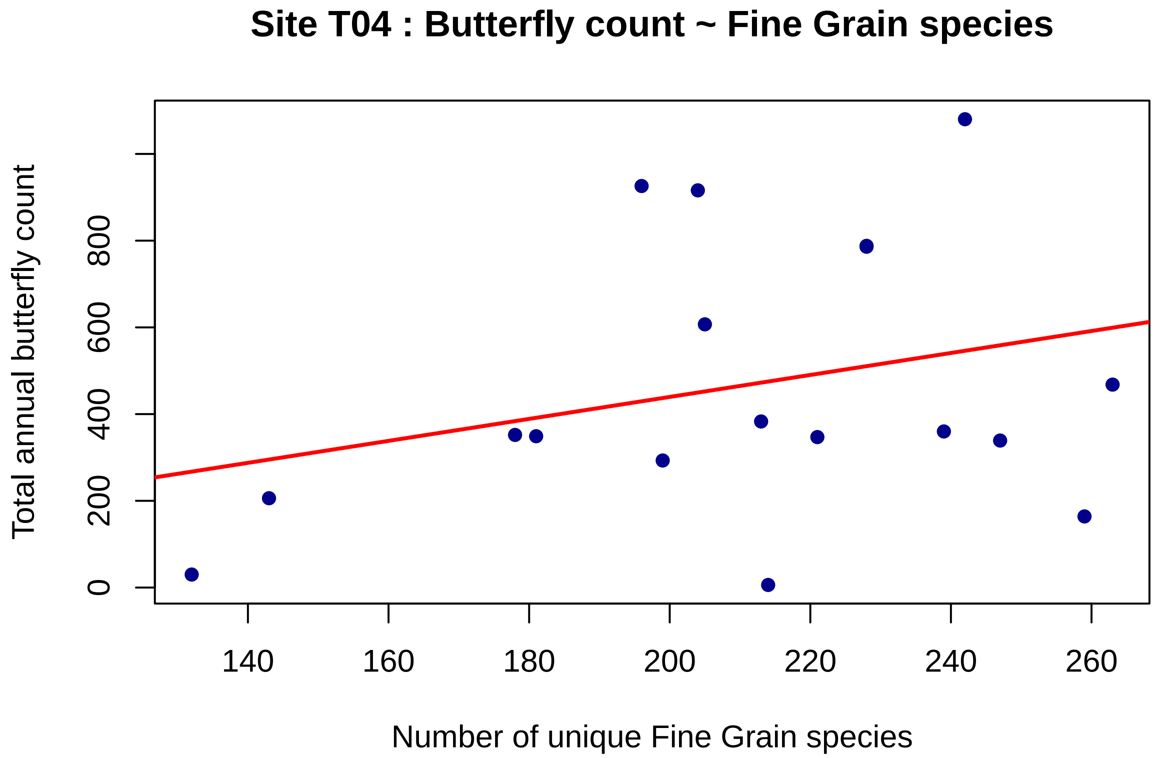


Table 6: Site T04 (n=18, R^2=0.084)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	-67.119	446.365	-0.150	8.82e-01
finegrain_species	2.534	2.090	1.212	2.43e-01

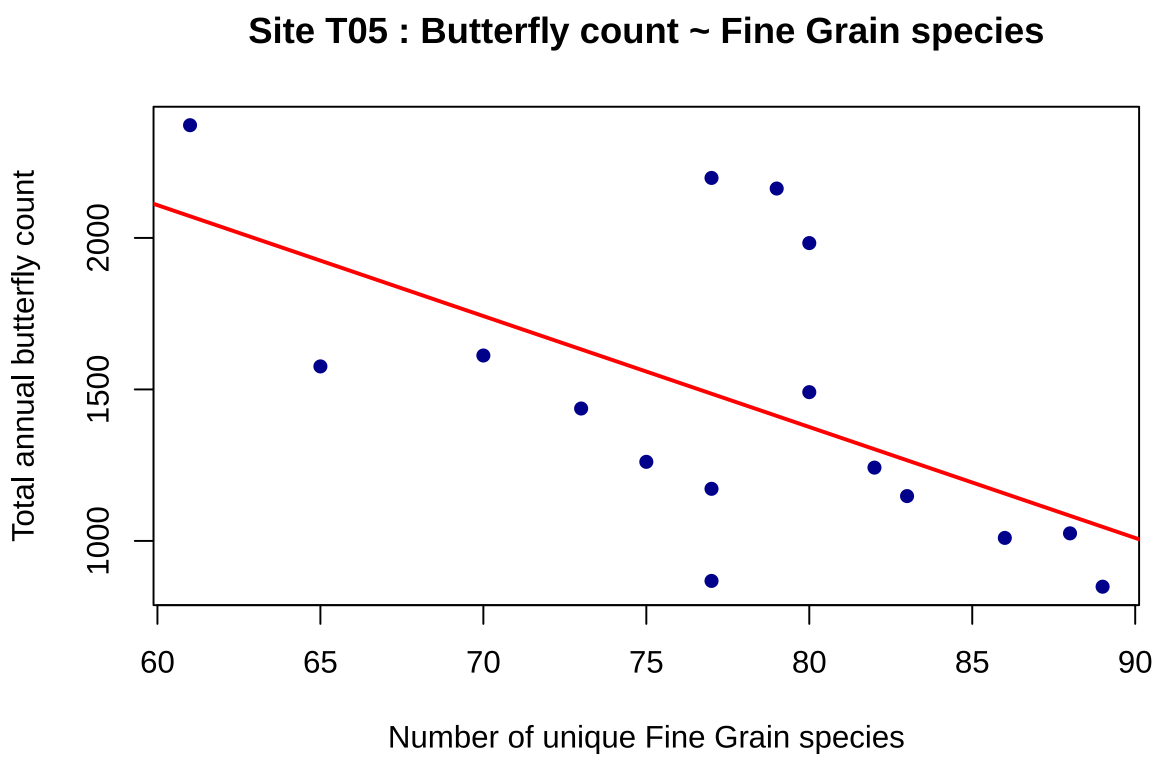


Table 7: Site T05 (n=16, R^2=0.335)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	4304.911	1074.379	4.007	1.30e-03
finegrain_species	-36.612	13.777	-2.657	1.88e-02

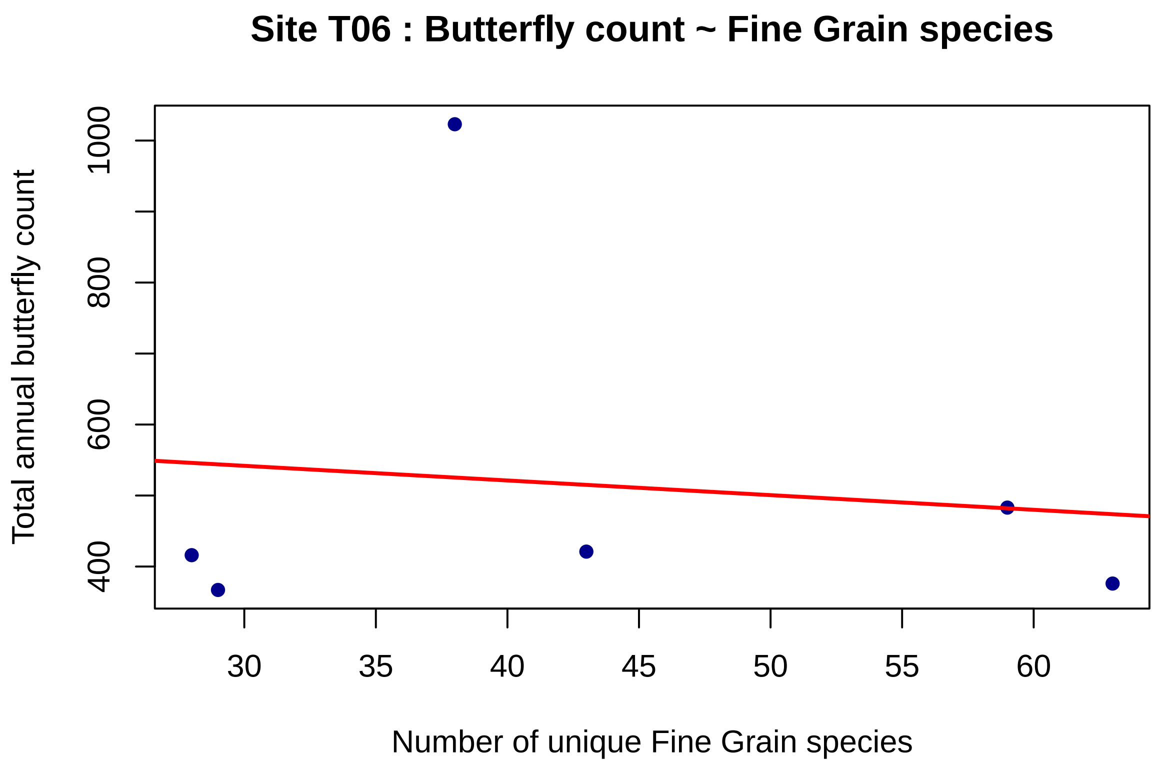


Table 8: Site T06 (n=6, R^2=0.015)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	603.794	383.464	1.575	1.90e-01
finegrain_species	-2.064	8.446	-0.244	8.19e-01

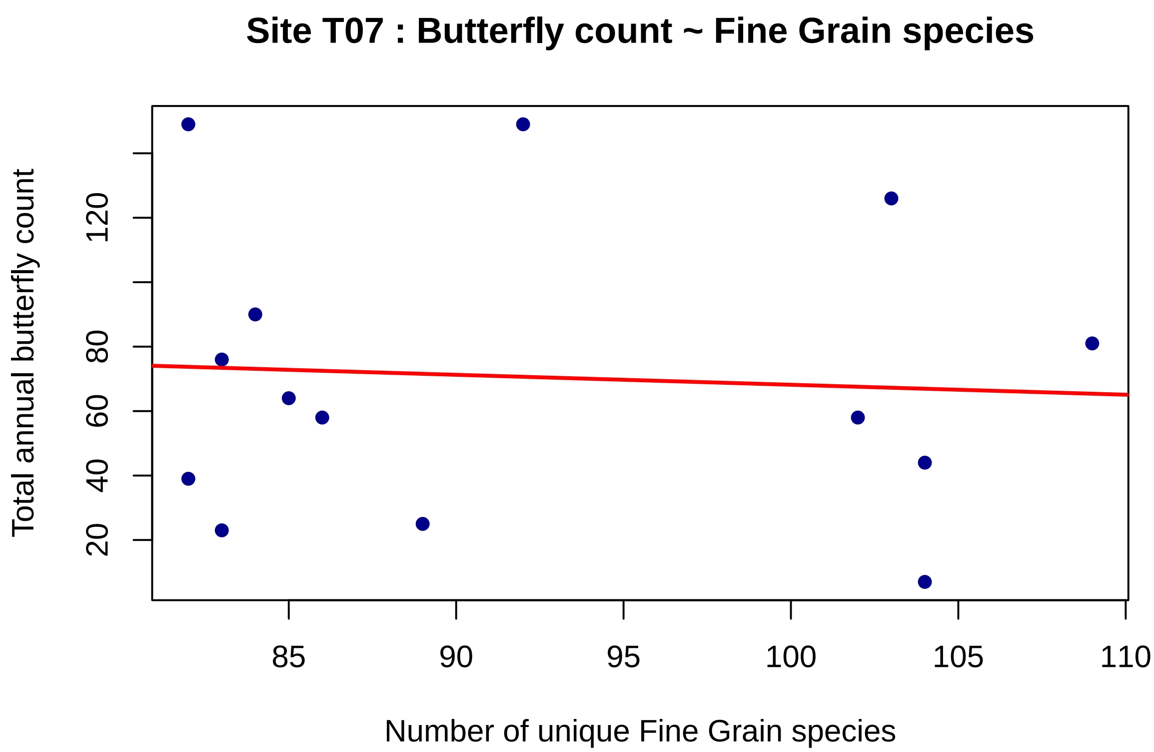


Table 9: Site T07 (n=14, R^2=0.005)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	99.053	119.084	0.832	4.22e-01
finegrain_species	-0.309	1.287	-0.240	8.14e-01

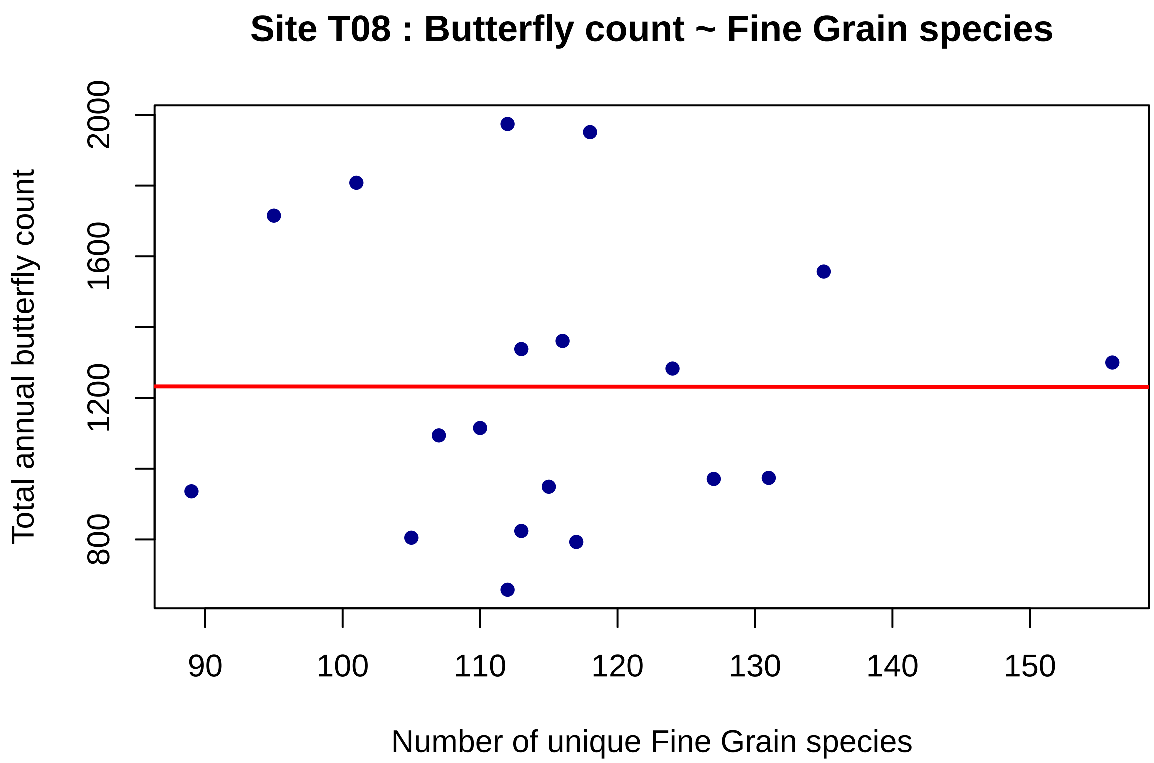


Table 10: Site T08 (n=19, R^2=0)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	1233.970	765.740	1.611	1.25e-01
finegrain_species	-0.018	6.573	-0.003	9.98e-01

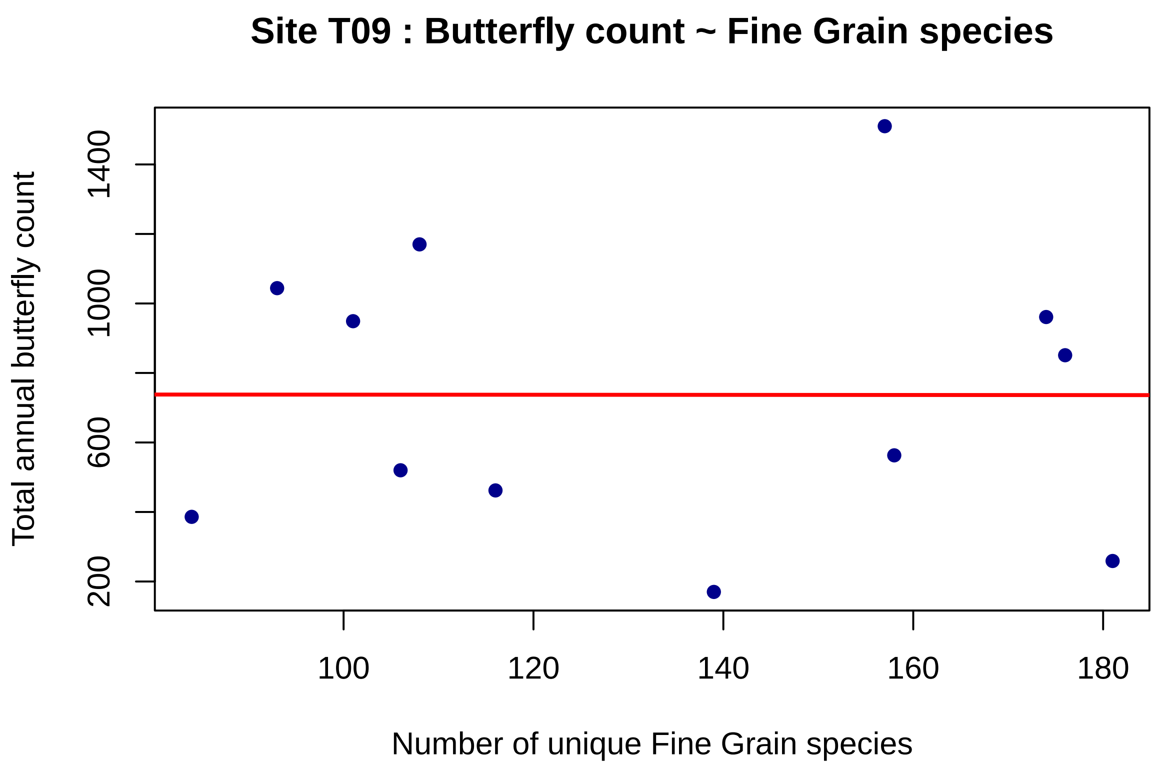


Table 11: Site T09 (n=12, R^2=0)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	739.149	497.738	1.485	1.68e-01
finegrain_species	-0.016	3.633	-0.004	9.97e-01

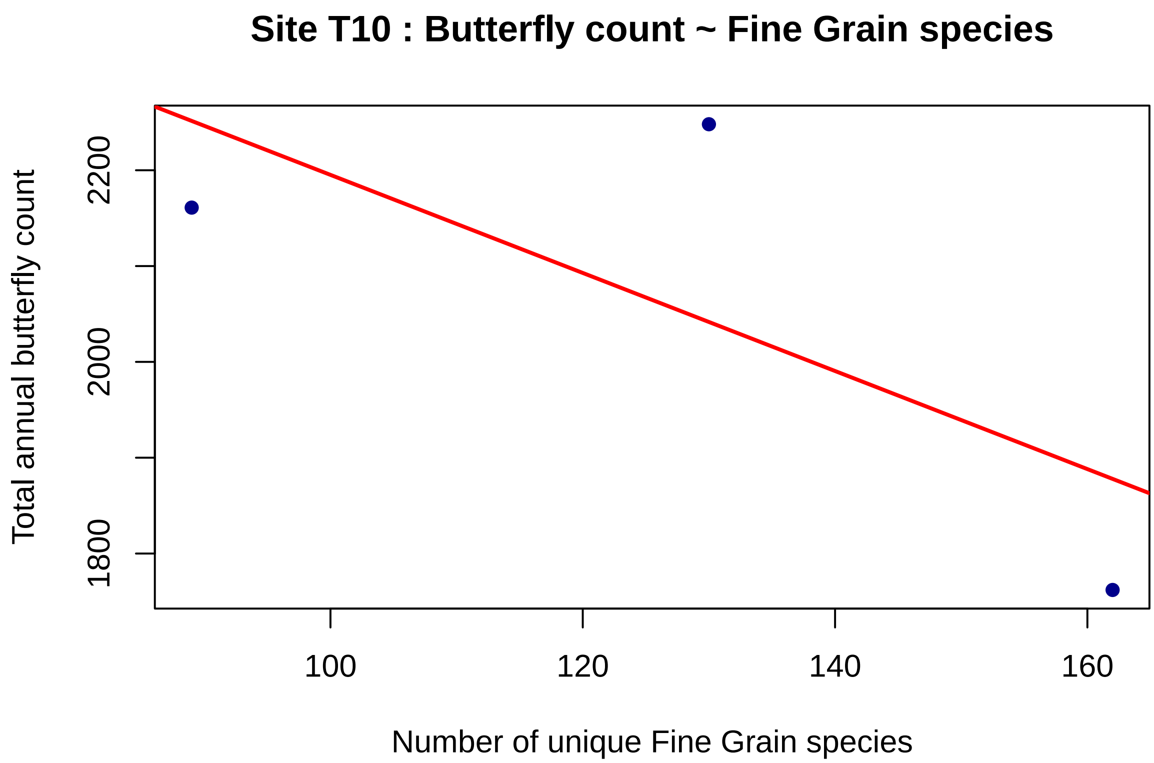


Table 12: Site T10 (n=3, R^2=0.522)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	2706.891	638.774	4.238	1.48e-01
finegrain_species	-5.117	4.896	-1.045	4.86e-01

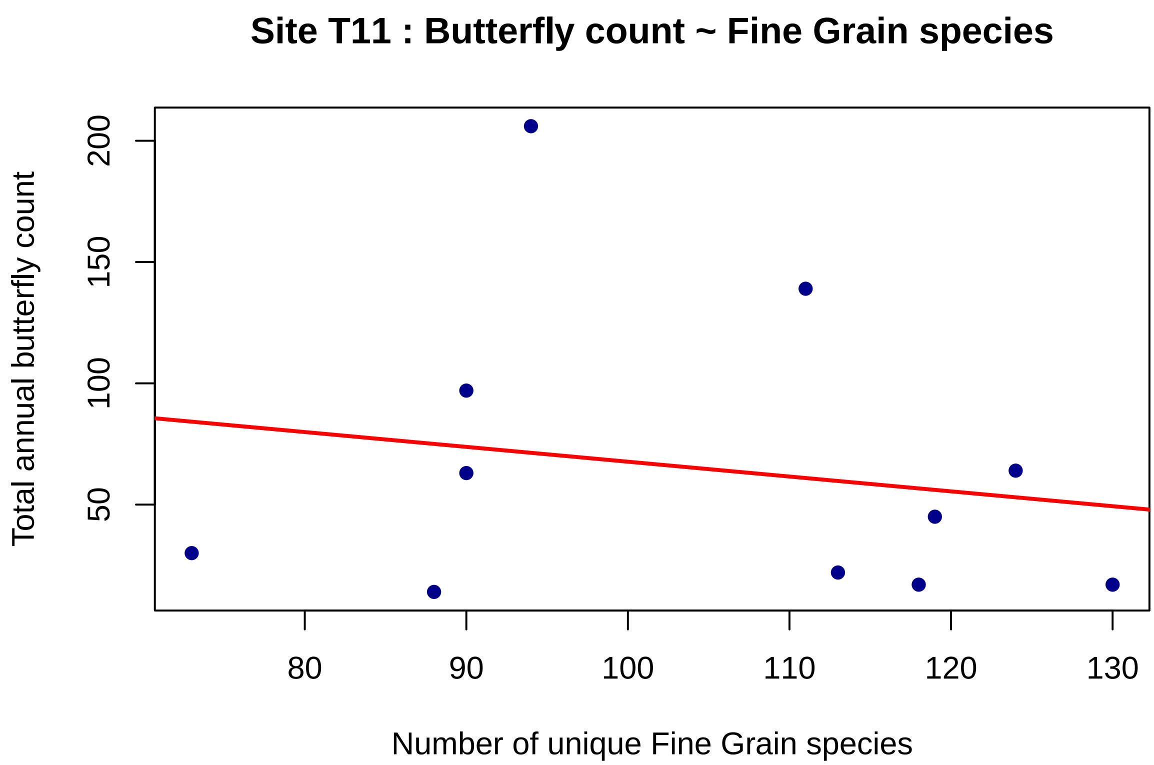


Table 13: Site T11 (n=11, R^2=0.034)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	128.823	115.826	1.112	2.95e-01
finegrain_species	-0.611	1.093	-0.559	5.90e-01

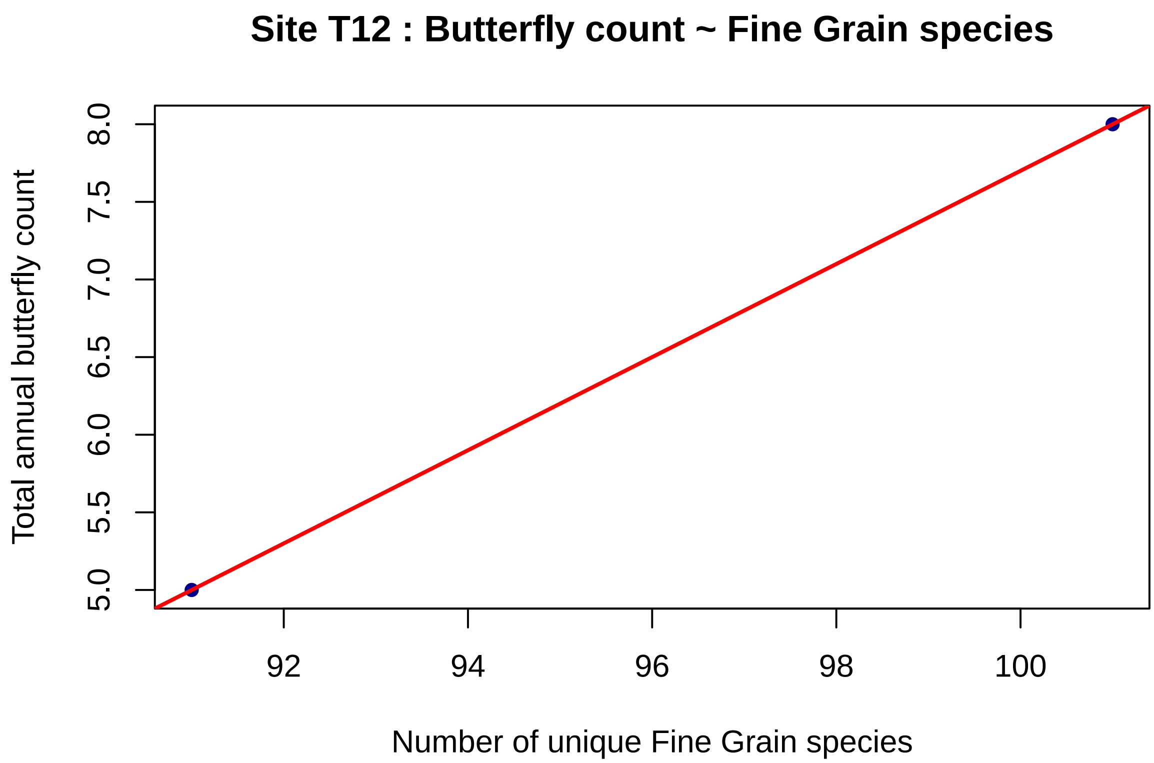


Table 14: Site T12 (n=2, R^2=1)

Term	Estimate	Std. Error	t value	Pr(>|t|)
Intercept	-22.3	NaN	NaN	NaN
finegrain_species	0.3	NaN	NaN	NaN

By fitting the same model to each site separately, we can better understand why the combined regression in part (c) is weak. Across the 12 sites, the slope varies from positive to nearly flat to negative and when combined, the data points scatter with no clear overall trend. Some sites have very few data points, leading to wide confidence intervals and unreliable slope estimates. Site-level models are more ecologically interpretable but come at the cost of statistical power due to smaller sample sizes.